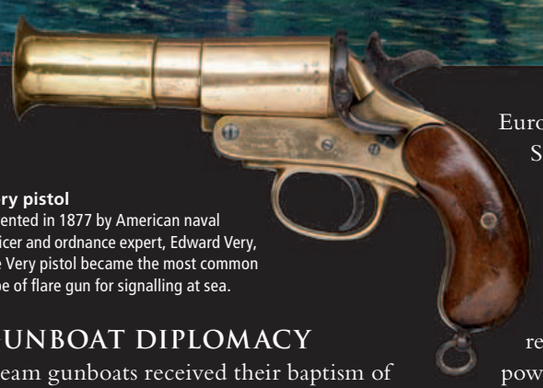


Civil war ironclads
The American Civil War was one of the first conflicts in which ironclad steamships went into battle. These Union gunboats and mortar boats are bombarding Confederate fortifications on the Mississippi.



Very pistol
Invented in 1877 by American naval officer and ordnance expert, Edward Very, the Very pistol became the most common type of flare gun for signalling at sea.

GUNBOAT DIPLOMACY

Steam gunboats received their baptism of fire in a scattering of conflicts around the world from the 1820s to the 1860s—from the Opium War through the Crimean War to the American Civil War, which exhibited the strengths and limitations of their varieties. A complex relationship developed between the evolution of these steam warships and the rampant imperialism of the 19th century. These new technologies increased the ability of the European powers, later joined by the United States and Japan, to bully less developed nations into submission. "Sending in the gunboat" became the answer to most popular assertions of anti-imperialism. But steamships also necessitated imperial conquest, as the ships' frequent need for refueling with coal required a global naval power to possess secure coaling stations around the world. Its later replacement with oil led Britain to put a high value upon controlling Iraqi oilfields during World War I.

RISING NAVAL POWERS

Growing rivalry between the imperialist powers gave the impetus to accelerating naval programs from the late 19th century. Up to that point Britain enjoyed unparalleled maritime dominance, not only through the Royal Navy but also through the overwhelming size of its merchant fleet and the output of its shipyards. By the 1890s, however, Britain's naval pre-eminence was being increasingly contested, even though world trade continued to be carried mostly in British ships. The influential writings of the American naval historian Alfred Mayer Mahan voiced a widespread view that a nation's global standing depended upon the strength of its navy. In an era of global competition, naval expansion became a popular cause promoted by jingoistic nationalists, even

1839
Start of First Opium War between Britain and China, an early instance of European "gunboat imperialism"

1843
Launch of *Great Britain*, first ocean-going screw-propelled ship

1855
French floating batteries bombard Kinburn; first ironclads used in battle



Case shot
This case or canister shot from the American Civil War, complete with small iron balls, was similar to grapeshot.

1871
Launch of first mastless turret ship, HMS *Devastation*

1830 | 1840 | 1850 | 1860 | 1870



Powder horn
This early 19th-century American powder horn was carried by marines and sailors for loading muskets and pistols.

1853
Start of Crimean War. First naval use of exploding shells at Sinope; race between shell and armor begins

1854
First strategic use of naval mines by Russians at Kronstadt

1859
France launches first seagoing ironclad warship *La Gloire*

1866
Battle of Lissa between Austria and Italy is first sea battle between armored ships